Linear Algebra — Lecture 7

LU Factorization, Determinants of Order 2, Determinants of Order n, Cofactor Expansion

Friedberg–Insel–Spence, *Linear Algebra* 4e, §4.1–§4.2
D. C. Lay, *Linear Algebra and Its Applications* 4e, §2.5, §3.1

Topics. The LU factorization and solving with L, U, permuted LU and permutation matrices, determinants of order 2 and their linearity in each row, the recursive definition of the determinant of order n, minors and cofactors, and cofactor expansion along any row, the determinants of triangular matrices, and how elementary row operations affect the determinant.

How the two texts divide the work. FIS (the main text) develops the theory of an **abstract vector space over an arbitrary field** $\mathbb{F}$; while Lay supplies concrete examples in $\mathbb{R}^n$ and the computational intuition. Mapping:

Topic	Lay 4e (printed)	Friedberg 4e (printed)
The LU factorization	§2.5 (p. 123–124)	—
Solving with L, U	§2.5 (p. 124–125)	—
The LU algorithm	§2.5 (p. 125–127)	—
Permuted LU, partial pivoting	§2.5 (p. 127)	—
Determinants of order 2	§3.1 (p. 164–165)	§4.1 (p. 199–201)
Order 3 and expansion	§3.1 (p. 165–166)	§4.2 (p. 209–211)
Linearity in each row	—	§4.2 (p. 212)
Expansion along any row	§3.1 (p. 166)	§4.2 (p. 215)
Triangular matrices and row operations	§3.1 (p. 167), §3.2 (p. 169–171)	§4.2 (p. 216)

Two questions that drive the whole lecture: (1) *Recording elimination as $A = LU$ (or $PA = LU$ when rows must be exchanged): which step does that save when solving, when inverting?* (2) *A square matrix is assigned a single number: by which rules is it pinned down, and which row should an expansion follow?*

How to read: for the LU section, use Lay first to get the algorithm under your fingers, then this section's theorems for $PA = LU$; for determinants, use Lay first to get the expansion of a 3×3 under your fingers, then Friedberg to see why the recursive definition and "expansion along any row" are correct.

All page numbers here are printed pages. Lay: PDF page = printed page +17. Friedberg: PDF page = printed page +15.

1 LU Factorization

Row reduction is the standard move for solving a system, and it has one drawback: every new right-hand side $\boldsymbol{b}$ forces the whole elimination again. Recording the **multipliers** used along the way produces a pair of triangular matrices L and U, after which every new right-hand side costs two substitutions. That is what a **matrix factorization** means: writing a matrix as a product of simpler matrices. Lay §2.5 (p. 123) calls it an analysis of the data, matching the synthesis that multiplication performs.

Definition 1.1 (LU factorization; Lay §2.5, p. 124)**.** Let A be an $m \times n$ matrix. If there is an $m \times m$ **unit lower triangular** matrix L (lower triangular with every diagonal entry equal to 1)

and an echelon form U of A such that

$$A = LU,$$

then this is an **LU factorization** of A.

Here U is only required to be *an* echelon form of A: it need not be square and need not have a pivot in every column. When A is $m \times n$, so is U, and its zero rows and its columns without a pivot stay where they are.

Theorem 1.1 (Row replacements produce an LU factorization; Lay §2.5, p. 125–126)**.** *Suppose A can be reduced to an echelon form U using only **row replacements**, that is, adding a multiple of one row to a row below it, and let $E_1, \dots, E_p$ be the elementary matrices of those replacements. Then*

$$L = (E_p \cdots E_1)^{-1}$$

is unit lower triangular, L is invertible, and $A = LU$.

Proof. Each row replacement is left multiplication by an elementary matrix (Lecture 5 notes, Theorem 3.1), so $E_p \cdots E_1 A = U$, that is, $A = (E_p \cdots E_1)^{-1} U$. The elementary matrix of a row replacement is itself unit lower triangular, and products and inverses of unit lower triangular matrices stay unit lower triangular (Lay §2.5, p. 126 and Exercise 19), so L is unit lower triangular; elementary matrices are invertible, so L is invertible. □

Proposition 1.2 (Solving a system with $A = LU$; Lay §2.5, p. 124)**.** *Let $A = LU$ be an LU factorization of A and let $\boldsymbol{b} \in \mathbb{F}^m$. Then $A\boldsymbol{x} = \boldsymbol{b}$ has the same solutions as the two triangular systems*

$$L\boldsymbol{y} = \boldsymbol{b}, \qquad U\boldsymbol{x} = \boldsymbol{y}.$$

The first is solved top-down (forward substitution), the second bottom-up (back substitution).

Proof. $A\boldsymbol{x} = L(U\boldsymbol{x})$. Put $\boldsymbol{y} = U\boldsymbol{x}$; then $A\boldsymbol{x} = \boldsymbol{b}$ is equivalent to "$L\boldsymbol{y} = \boldsymbol{b}$ and $U\boldsymbol{x} = \boldsymbol{y}$". Conversely, if $\boldsymbol{y}$ satisfies $L\boldsymbol{y} = \boldsymbol{b}$ and $\boldsymbol{x}$ satisfies $U\boldsymbol{x} = \boldsymbol{y}$, then $A\boldsymbol{x} = L\boldsymbol{y} = \boldsymbol{b}$. Since L is unit lower triangular, $L\boldsymbol{y} = \boldsymbol{b}$ has, for every right-hand side $\boldsymbol{b}$, a unique solution $\boldsymbol{y}$ found from the top (step i involves only y_i, with coefficient 1); hence L is invertible. □

Algorithm 1.1 (LU factorization; Lay §2.5, p. 126)**.**

Input: $A \in M_{m\times n}(\mathbb{F})$ *that can be reduced to an echelon form without interchanging rows.*

Output: a unit lower triangular L and an echelon form U with $A = LU$.

1. *set $U = A$ and $L = I_m$;*
2. *for $j = 1, 2, \dots, \min(m, n)$:*
   (a) *if rows j to m of column j are all zero, the column has no pivot: skip it;*
   (b) *otherwise, for each $i > j$ put $c = U_{ij}/U_{jj}$, replace R_i by $R_i - cR_j$ and write c into L_{ij}.*
3. *output L and U.*

Proof. Step 2(b) rests on the definition $L = (E_p \cdots E_1)^{-1}$: left multiplication by $E_p \cdots E_1$ is what carries L back to I, so the very sequence of row replacements that carries A to U carries L to I, and the (i, j)-entry of L is that multiplier c: the elementary matrix of $R_i \leftarrow R_i - cR_j$ is $I - c\,e_i e_j^\top$, whose inverse $I + c\,e_i e_j^\top$ puts c into L. □

Example 1.1 (An LU factorization of a 4×5 matrix; Lay Example 2, §2.5, p. 126–127)**.** Find an LU factorization of

$$A = \begin{bmatrix} 2 & 4 & -1 & 5 & -2 \\ -4 & -5 & 3 & -8 & 1 \\ 2 & -5 & -4 & 1 & 8 \\ -6 & 0 & 7 & -3 & 1 \end{bmatrix}$$

Solution. Row 1 clears column 1 (add 2 times row 1 to row 2, subtract row 1 from row 3, add 3 times row 1 to row 4); row 2 then clears column 2 (add 3 times to row 3, subtract 4 times from row 4); column 3 has no pivot, so move to column 4 (subtract 2 times row 3 from row 4). This gives

$$A \sim \begin{bmatrix} 2 & 4 & -1 & 5 & -2 \\ 0 & 3 & 1 & 2 & -3 \\ 0 & -9 & -3 & -4 & 10 \\ 0 & 12 & 4 & 12 & -5 \end{bmatrix} \sim \begin{bmatrix} 2 & 4 & -1 & 5 & -2 \\ 0 & 3 & 1 & 2 & -3 \\ 0 & 0 & 0 & 2 & 1 \\ 0 & 0 & 0 & 4 & 7 \end{bmatrix}$$

$$\sim U = \begin{bmatrix} 2 & 4 & -1 & 5 & -2 \\ 0 & 3 & 1 & 2 & -3 \\ 0 & 0 & 0 & 2 & 1 \\ 0 & 0 & 0 & 0 & 5 \end{bmatrix}.$$

Recorded as the added multiple c, the multipliers are $c = 2, -1, 3$ (column 1, rows $2, 3, 4$), then $c = 3, -4$ (column 2, rows $3, 4$), then $c = -2$ (column 4, row 4); with the sign changed they fill L:

$$L = \begin{bmatrix} 1 & 0 & 0 & 0 \\ -2 & 1 & 0 & 0 \\ 1 & -3 & 1 & 0 \\ -3 & 4 & 2 & 1 \end{bmatrix}$$

Check: the second row of LU is $2 \cdot (2, 4, -1, 5, -2) + (0, 3, 1, 2, -3) = (-4, -5, 3, -8, 1)$, which is row 2 of A; the other three rows match the same way. □

Approach. Eliminate column by column from the left, and after clearing each column put the multipliers with the sign changed into the corresponding column of L; U is the echelon form that results. A has four rows, so L is 4×4. If some column has no pivot, skip it and keep that pivotless column in U.

Remark 1.1 (Permuted LU and partial pivoting; Lay §2.5, p. 127)**.** In practice row interchanges are almost always needed: partial pivoting takes, as the pivot, the entry of largest absolute value in the column, so that no division by a tiny number occurs. Once interchanges are allowed, L is no longer lower triangular but **permuted lower triangular**: rearranging the rows of L in the order the pivots appear turns it back into a unit lower triangular matrix. Solving works as before, except that forward substitution follows the pivot order from left to right. The phrase "LU factorization" usually includes this permuted case.

Definition 1.2 (Permutation matrix)**.** An $n \times n$ matrix is a **permutation matrix** if every row and every column contains exactly one 1 and all other entries are 0.

Lemma 1.3 (Properties of a permutation matrix)**.** *Let P be an $n \times n$ permutation matrix. Then*

(a) P is invertible, with $P^{-1} = P^{\top}$;
(b) $P^{\top}$ is again a permutation matrix;
(c) left multiplication by P reorders the rows of A in the order fixed by P, and right multiplication by $P^{\top}$ reorders its columns.

Proof.

(a) Writing $\sigma(i)$ for the column holding the 1 of row i, each column has a single 1, so σ is an injection of $\{1,\dots,n\}$ into itself, that is, a permutation. The (i,j)-entry of $PP^\top$ is $\sum_k P_{ik}P_{jk}$, which equals 1 only when $\sigma(i)=\sigma(j)$ and therefore, σ being injective, only when $i=j$; hence $PP^\top = I_n$. The same argument gives $P^\top P = I_n$.

(b) Transposition turns rows into columns and columns into rows, so $P^\top$ again has exactly one 1 in every row and every column.

(c) Row i of P is $\boldsymbol{e}_{\sigma(i)}^\top$, so row i of PA is row $\sigma(i)$ of A. Since $(AP^\top)^\top = PA^\top$, right multiplication by $P^\top$ reorders the columns in the same order. □

Example 1.2 (A row permutation composed with elimination, by blocks)**.** Let

$$P = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}, \qquad E = \begin{bmatrix} 1 & 0 & 0 \\ a & 1 & 0 \\ b & 0 & 1 \end{bmatrix},$$

where P interchanges rows 2 and 3 and E is the elimination matrix of the first step, with multipliers a and b. Split as 1+2: $P = \begin{bmatrix} 1 & \mathbf{0}^\top \\ \mathbf{0} & S \end{bmatrix}$, $E = \begin{bmatrix} 1 & \mathbf{0}^\top \\ \boldsymbol{m} & I_2 \end{bmatrix}$, where $S = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$ and $\boldsymbol{m} = (a,b)^\top$. Compute PEP^{-1} by blocks.

Solution. $P^{-1} = P^\top = P$. First PE: the top-left block is $1 \cdot 1 = 1$, the top-right block is $1 \cdot \mathbf{0}^\top = \mathbf{0}^\top$, the bottom-left block is $S\boldsymbol{m} = (b,a)^\top$, and the bottom-right block is $SI_2 = S$, so $PE = \begin{bmatrix} 1 & \mathbf{0}^\top \\ S\boldsymbol{m} & S \end{bmatrix}$. Multiplying on the right by $P^{-1} = \begin{bmatrix} 1 & \mathbf{0}^\top \\ \mathbf{0} & S \end{bmatrix}$ leaves the bottom-left block alone and turns the bottom-right one into $S^2 = I_2$, hence

$$PEP^{-1} = \begin{bmatrix} 1 & \mathbf{0}^\top \\ S\boldsymbol{m} & I_2 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ b & 1 & 0 \\ a & 0 & 1 \end{bmatrix}.$$

The shape is that of E; only the two multipliers have changed places. □

Approach. PEP^{-1} means "permute the rows by P, eliminate, then permute back", and on an elimination matrix it only reorders the multiplier column; the upper triangle stays zero. Theorem 1.5 below is the general form of this observation.

Lemma 1.4 (A permutation only reorders the multipliers)**.** *Let $P = \begin{bmatrix} I_j & \mathbf{0} \\ \mathbf{0} & S \end{bmatrix}$, where S is a permutation matrix of size $n-j$, so that P only reorders the rows below row j. Let the elimination matrix of step j be $E = \begin{bmatrix} E_1 & \mathbf{0} \\ A_1 & I_{n-j} \end{bmatrix}$, with E_1 a unit lower triangular $j \times j$ matrix and A_1 the block of multipliers. Then*

$$PEP^{-1} = \begin{bmatrix} E_1 & \mathbf{0} \\ SA_1 & I_{n-j} \end{bmatrix},$$

again unit lower triangular of the same shape: the top-left block is untouched and the multiplier block is reordered by S.

Proof. Multiply blockwise in the split j and $n-j$: $PE = \begin{bmatrix} E_1 & \mathbf{0} \\ SA_1 & S \end{bmatrix}$, and right multiplication by $P^{-1} = \begin{bmatrix} I_j & \mathbf{0} \\ \mathbf{0} & S^{-1} \end{bmatrix}$ turns the bottom-right block into $SS^{-1} = I_{n-j}$. The upper-right block is zero and the diagonal entries are 1, so the result is unit lower triangular. □

Theorem 1.5 (LU factorization with permutations; Lay §2.5, p. 127)**.** *For every $A \in M_{m\times n}(\mathbb{F})$ there are a permutation matrix P, a unit lower triangular matrix L and an echelon form U such that*

$$PA = LU.$$

Proof. Induction on the number of columns n. For $n = 1$ a permutation moves a nonzero entry to the first row (if the column is zero, take $U = 0$) and $L = 1$.

Let $n \geq 2$. Pick a nonzero entry in column 1, move it to position $(1, 1)$ by a permutation P_1, and clear the rest of column 1 with an elimination matrix E_1:

$$E_1 P_1 A = \begin{bmatrix} a & \boldsymbol{r}^\top \\ \mathbf{0} & A' \end{bmatrix}, \qquad A' \in M_{(m-1)\times(n-1)}(\mathbb{F}).$$

By the induction hypothesis there is a permutation P' with $P'A' = L'U'$. Put $Q = \begin{bmatrix} 1 & \mathbf{0}^\top \\ \mathbf{0} & P' \end{bmatrix}$ and multiply on the left:

$$QE_1P_1A = \begin{bmatrix} a & \boldsymbol{r}^\top \\ \mathbf{0} & L'U' \end{bmatrix} = \begin{bmatrix} 1 & \mathbf{0}^\top \\ \mathbf{0} & L' \end{bmatrix} \begin{bmatrix} a & \boldsymbol{r}^\top \\ \mathbf{0} & U' \end{bmatrix}.$$

Call the two factors on the right D and U. Both Q and D act only below row 2 and are block diagonal, so they commute; and by Lemma 1.4, $QE_1^{-1}Q^{-1}$ is again unit lower triangular. Hence

$$PA = LU, \qquad P = QP_1, \quad L = \big(QE_1^{-1}Q^{-1}\big)D, \quad U = \begin{bmatrix} a & \boldsymbol{r}^\top \\ \mathbf{0} & U' \end{bmatrix},$$

and L, a product of two unit lower triangular matrices, is unit lower triangular. □

Algorithm 1.2 (LU factorization with row interchanges (partial pivoting))**.**

Input: $A \in M_{m\times n}(\mathbb{F})$.
Output: *a permutation matrix P, a unit lower triangular L and an echelon form U with $PA = LU$.*

1. *set $U = A$, $L = I_m$ and $P = I_m$;*
2. *For $j = 1, 2, \ldots, \min(m, n)$:*
   (a) *find the entry of largest absolute value in rows j to m of column j; if they are all zero, the column has no pivot: skip it and keep the row index unchanged;*
   (b) *if that largest entry is not in row j (say it is in row r), interchange rows j and r of U, interchange rows j and r of P, and interchange rows j and r of the **already filled** first $j - 1$ columns of L;*
   (c) *for each $i > j$: put $c = U_{ij}/U_{jj}$, replace R_i by $R_i - cR_j$, write c into L_{ij}.*
3. *Output P, L, U.*

Proof. Every row replacement is left multiplication by a unit lower triangular elementary matrix, and every interchange is left multiplication by a permutation matrix; permuting the already filled columns of L at the same moment is a conjugation of the recorded multipliers, and by Lemma 1.4 the shape is preserved. The permutation chosen in step 2(a) only reorders rows below row j, which is exactly the hypothesis used in the proof of Theorem 1.5, so at the end $LU = PA$. Substituting the result back into $LU = PA$ also detects whether the multipliers were carried along in an interchange. □

Remark 1.2 (The interleaved product and the reordered multipliers)**.** Written out in order, the elimination and the permutations give $U = E_m P_m \cdots E_1 P_1 A$. The proof of Theorem 1.5 is exactly the process of absorbing each permutation into $\tilde{E}_k = P_k^{-1} E_k P_k$: by Lemma 1.4 this is again unit lower triangular, with the multipliers reordered by P_k. The permutations thus move left one by one until a single P and a single unit lower triangular L remain, that is $LU = PA$, equivalently

$A = P^\top LU$. Computation need not perform these conjugations; how the bookkeeping is done in practice is Algorithm 1.2.

Example 1.3 (An LU factorization that needs a row interchange)**.** Find the permuted LU factorization of $A = \begin{bmatrix} 0 & 1 & 2 \\ 1 & 2 & 3 \\ 1 & 3 & 1 \end{bmatrix}$.

Solution. The nonzero entries of column 1 lie in rows 2 and 3, so interchange rows 1 and 2:

$$P_1 = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \qquad P_1 A = \begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & 2 \\ 1 & 3 & 1 \end{bmatrix}.$$

Clear the first entry of row 3 with the first row, the multiplier being 1:

$$E_1 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -1 & 0 & 1 \end{bmatrix}, \qquad E_1 P_1 A = \begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & 2 \\ 0 & 1 & -2 \end{bmatrix}.$$

The pivot in column 2 is 1, so no interchange there; subtract row 2 from row 3. The multipliers go into L by position: 0 at $(2,1)$ (after the interchange the leading entry of that row is already zero), and 1 at $(3,1)$ and $(3,2)$. Hence

$$U = \begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & 2 \\ 0 & 0 & -4 \end{bmatrix}, \qquad L = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 1 & 1 & 1 \end{bmatrix}, \qquad P = P_1.$$

Check: $LU = P_1 A$, row by row. □

Approach. Choose the pivot as the entry of largest absolute value in its column; that interchange happens between rows 1 and 2, and no further interchange occurs, so $P = P_1$. The multipliers go into L by position, because at the time of the interchange L held no multiplier yet.

Remark 1.3 (Why LU rather than A^{-1}; Lay §2.5, p. 127)**.** An LU factorization costs about $2n^3/3$ multiplications and additions, the same order as one elimination; afterwards each new right-hand side costs $O(n^2)$ for the two substitutions. Solving through A^{-1} first pays $O(n^3)$ to produce A^{-1}, and when A is sparse A^{-1} is usually dense, so multiplying by it is slow. For a sequence of right-hand sides with the same A, the difference is the difference between "factor once, substitute many times" and "invert again for every right-hand side".

Remark 1.4 (A whole A^{-1} from one LU; Lay §2.5, p. 127)**.** The j-th column of A^{-1} is the solution of $A\boldsymbol{x} = \boldsymbol{e}_j$. So once $A = LU$ is known, computing the whole inverse means taking $\boldsymbol{e}_1, \ldots, \boldsymbol{e}_n$ as right-hand sides in turn, each with one forward and one back substitution: factor once, substitute n times. How this compares in cost with "compute A^{-1} and multiply" is Remark 1.3.

2 Determinants of Order 2

The first lectures treated a matrix as a linear transformation and as a system of equations; the matrix itself stayed a table of numbers. From this lecture on, each square matrix receives a single number. Two rules characterise it: fixing the other rows, it is linear in one row, and it changes sign when two rows are interchanged. In order 2 the formula can be written down at once; in order n the determinant is defined recursively through cofactors.

Definition 2.1 (Determinant of order 2; Friedberg §4.1, p. 200)**.** Let $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ be a 2×2 matrix with entries from a field $\mathbb{F}$. The **determinant** of A, written $\det A$ or $|A|$, is the scalar

$$ad - bc.$$

Example 2.1 (Friedberg Example 1, §4.1, p. 200)**.** Find the determinants of $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$ and $B = \begin{bmatrix} 3 & 2 \\ 6 & 4 \end{bmatrix}$.

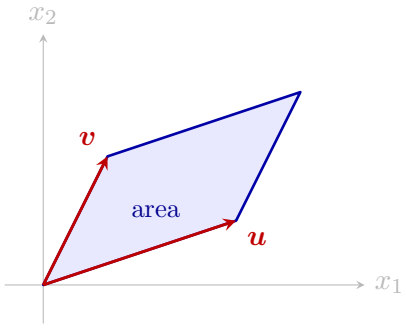
Solution. Both follow the definition:

$$\det A = 1 \cdot 4 - 2 \cdot 3 = -2, \qquad \det B = 3 \cdot 4 - 2 \cdot 6 = 0.$$

In B the second row is twice the first, so the two products agree and the determinant vanishes. □

Approach. Order 2 has a single formula: subtract the product of one diagonal from the product of the other. Proportional rows force the value 0, which is the rank criterion of Lecture 4 seen from the determinant side.

Remark 2.1 (Geometric picture: the area of a parallelogram)**.** Let $A = (\boldsymbol{u}\ \boldsymbol{v})$ be a 2×2 matrix with columns $\boldsymbol{u}$ and $\boldsymbol{v}$. Then $|\det A|$ is the area of the parallelogram with sides $\boldsymbol{u}$ and $\boldsymbol{v}$, and the sign of $\det A$ records the orientation: positive when $\boldsymbol{u}$ turns counterclockwise into $\boldsymbol{v}$, negative otherwise.



The picture takes $\boldsymbol{u} = (3, 1)$ and $\boldsymbol{v} = (1, 2)$, so $\det A = 3 \cdot 2 - 1 \cdot 1 = 5$ and the area of the parallelogram is indeed 5. Interchanging the two columns keeps the area and flips the sign, a consequence of linearity in each row: two equal columns give area 0, and the determinant 0 as well.

Theorem 2.1 (Linearity in each row; Friedberg Theorem 4.1, §4.1, p. 200)**.** *Let $\boldsymbol{u}, \boldsymbol{v}, \boldsymbol{w} \in \mathbb{F}^2$ and $k \in \mathbb{F}$. With the other row held fixed, the determinant is a linear function of this row:*

$$\det \begin{bmatrix} \boldsymbol{u} + k\boldsymbol{v} \\ \boldsymbol{w} \end{bmatrix} = \det \begin{bmatrix} \boldsymbol{u} \\ \boldsymbol{w} \end{bmatrix} + k \det \begin{bmatrix} \boldsymbol{v} \\ \boldsymbol{w} \end{bmatrix}, \qquad \det \begin{bmatrix} \boldsymbol{w} \\ \boldsymbol{u} + k\boldsymbol{v} \end{bmatrix} = \det \begin{bmatrix} \boldsymbol{w} \\ \boldsymbol{u} \end{bmatrix} + k \det \begin{bmatrix} \boldsymbol{w} \\ \boldsymbol{v} \end{bmatrix}.$$

Proof. Write $\boldsymbol{u} = (a_1, a_2)$, $\boldsymbol{v} = (b_1, b_2)$, $\boldsymbol{w} = (c_1, c_2)$. By definition the left side is

$$(a_1 + kb_1)c_2 - (a_2 + kb_2)c_1 = (a_1c_2 - a_2c_1) + k(b_1c_2 - b_2c_1),$$

which is the sum of the two determinants on the right. The second identity follows the same way with c moved to the top row. □

Proposition 2.2 (Two rows are dependent exactly when the determinant vanishes)**.** *Let* $\boldsymbol{u}, \boldsymbol{w} \in \mathbb{F}^2$*. Then* $\det \begin{bmatrix} \boldsymbol{u} \\ \boldsymbol{w} \end{bmatrix} = 0$ *exactly when* $\boldsymbol{u}$ *and* $\boldsymbol{w}$ *are linearly dependent.*

Proof. If $\boldsymbol{w} = \mathbf{0}$ both rows are the zero vector, they are dependent, and the determinant is 0. Let $\boldsymbol{w} = (w_1, w_2) \neq \mathbf{0}$. By Theorem 2.1, $f(\boldsymbol{u}) = \det \begin{bmatrix} \boldsymbol{u} \\ \boldsymbol{w} \end{bmatrix}$ is a linear functional on $\mathbb{F}^2$. If $w_2 \neq 0$, the equation $u_1 w_2 - u_2 w_1 = 0$ gives $\boldsymbol{u} = \dfrac{u_2}{w_2}\boldsymbol{w}$; if $w_2 = 0$ then $w_1 \neq 0$ and the equation gives $u_2 = 0$. Either way the zeros of f are exactly the multiples of $\boldsymbol{w}$. Conversely, if $\boldsymbol{u}$ is a multiple of $\boldsymbol{w}$ the two rows are proportional and the determinant is 0. □

Remark 2.2 (det is not a linear transformation; Friedberg §4.1, p. 200)**.** The two matrices of Example 2.1 add up to $A + B = \begin{bmatrix} 4 & 4 \\ 9 & 8 \end{bmatrix}$, so $\det(A+B) = 4 \cdot 8 - 4 \cdot 9 = -4$ while $\det A + \det B = -2 + 0 = -2$. The two differ, so $\det : M_{2\times 2}(\mathbb{F}) \to \mathbb{F}$ is not a linear map: it is linear along each row separately, not on matrices as a whole.

Remark 2.3 (The invertibility criterion of order 2)**.** Lecture 4 notes, Theorem 3.1 already states that $\det A \neq 0$ exactly when A is invertible, and gives $A^{-1} = \frac{1}{\det A} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$. Friedberg Theorem 4.2 (§4.1, p. 201) says the same thing with a rank argument: when $\det A \neq 0$ one verifies directly that the displayed matrix is the inverse; when $\det A = 0$ the rank of A is at most 1.

3 Determinants of Order n and Cofactor Expansion

The formula for order 2 has no ready-made counterpart in order n, but deleting one row and one column leaves a smaller square matrix, and its determinant can be taken in turn. The determinant is therefore defined by recursion.

Definition 3.1 (Minor, cofactor, determinant of order n; Friedberg §4.2, p. 209–210)**.** Let $A \in M_{n\times n}(\mathbb{F})$ with $n \geq 2$. Deleting row i and column j of A leaves an $(n-1)\times(n-1)$ matrix, written $\tilde{A}_{ij}$ and called the **minor** of A_{ij}. The scalar

$$c_{ij} = (-1)^{i+j} \det(\tilde{A}_{ij})$$

is the **cofactor** of A_{ij}. The determinant of A is defined recursively by **cofactor expansion along the first row**:

$$\det A = \sum_{j=1}^{n} (-1)^{1+j} A_{1j} \det(\tilde{A}_{1j}) = A_{11}c_{11} + A_{12}c_{12} + \cdots + A_{1n}c_{1n}.$$

Example 3.1 (Friedberg §4.2, Example 1, p. 210)**.** Find the determinant of $A = \begin{bmatrix} 1 & 3 & -3 \\ -3 & -5 & 2 \\ -4 & 4 & -6 \end{bmatrix}$.

Solution. Expand along the first row; the three minors are

$$\det \begin{bmatrix} -5 & 2 \\ 4 & -6 \end{bmatrix} = 30 - 8 = 22, \quad \det \begin{bmatrix} -3 & 2 \\ -4 & -6 \end{bmatrix} = 18 + 8 = 26, \quad \det \begin{bmatrix} -3 & -5 \\ -4 & 4 \end{bmatrix} = -12 - 20 = -32.$$

The signs alternate as $(-1)^{1+j}$, hence

$$\det A = 1 \cdot 22 - 3 \cdot 26 + (-3) \cdot (-32) = 22 - 78 + 96 = 40.$$ □

Approach. First turn each minor into a single number, then combine them with alternating signs. A sign depends only on the position (i, j), never on the sign of the entry itself; here $(-1)^{1+j}$ gives the checkerboard pattern plus, minus, plus.

Remark 3.1 (The $n!$-term formula and inversions)**.** Unfolding the cofactor expansion layer by layer writes the determinant as a sum of $n!$ terms. Let σ be a permutation of $\{1, \dots, n\}$ and let $\tau(\sigma)$ be its **number of inversions**, the count of pairs $i < j$ with $\sigma(i) > \sigma(j)$; put $\operatorname{sgn}(\sigma) = (-1)^{\tau(\sigma)}$. Then

$$\det A = \sum_{\sigma} \operatorname{sgn}(\sigma)\, A_{1\sigma(1)} A_{2\sigma(2)} \cdots A_{n\sigma(n)},$$

the sum running over all $n!$ permutations. For $n = 2$ the two terms are $A_{11}A_{22} - A_{12}A_{21}$, which agrees with Definition 2.1; for $n = 3$ the six terms combine into the expansion one remembers by drawing diagonals.

This formula and cofactor expansion produce the same number: each step of the recursion takes one entry from a row and deletes its column, and after n steps what has been taken is exactly a permutation; the sign is the parity of the number of interchanges needed to sort that permutation back to $1, 2, \dots, n$, which is $(-1)^{\tau(\sigma)}$.

Neither textbook adopts this definition. Friedberg defines the determinant recursively through cofactors (§4.2, p. 209–210) and then, in §4.5* (from p. 238), characterises it as the unique function that is linear in each row and vanishes on matrices with two equal rows; Lay expands along a row (§3.1, p. 165–166). The inversion formula is equivalent to those characterisations and merely writes the unfolding out as an explicit sum.

Theorem 3.1 (Linearity in each row; Friedberg Theorem 4.3, §4.2, p. 212)**.** *Let $A \in M_{n\times n}(\mathbb{F})$. If row r is replaced by $\boldsymbol{u} + k\boldsymbol{v}$ while the other rows stay fixed, the determinant equals the sum of the determinants obtained by putting $\boldsymbol{u}$ and $\boldsymbol{v}$ in row r, the second weighted by k.*

Proof. Induction on n. For $n = 1$ the determinant is the entry itself, so the claim is clear. Let $n \geq 2$ and assume the result for order $n - 1$.

If $r = 1$: by definition $\det A = \sum_j A_{1j} c_{1j}$, and the cofactor c_{1j} involves only rows 2 to n, so the sum is linear in row 1.

If $r > 1$: expand along the first row, $\det A = \sum_j A_{1j} c_{1j}$ with $c_{1j} = \pm \det(\tilde{A}_{1j})$. Row $r - 1$ of $\tilde{A}_{1j}$ is row r of A and its other rows agree with those of A; by the induction hypothesis each $\det(\tilde{A}_{1j})$ is linear in that row, so the weighted sum of these determinants of order $n - 1$ is linear in row r of A. □

Lemma 3.2 (Friedberg §4.2, p. 213)**.** *Let $B \in M_{n\times n}(\mathbb{F})$ with $n \geq 2$. If row i of B is $\boldsymbol{e}_k$, then*

$$\det B = (-1)^{i+k} \det(\tilde{B}_{ik}).$$

Proof. For $i = 1$ the definition gives the claim at once: the first row has 1 only in position k, so a single term survives, $(-1)^{1+k} \cdot 1 \cdot \det(\tilde{B}_{1k})$. The case $i > 1$ is proved by induction on n: expand B along its first row and apply the induction hypothesis to each minor of order $n - 1$; the signs combine to $(-1)^{i+k}$. The step-by-step details are in Friedberg §4.2, p. 213–215. □

Theorem 3.3 (Expansion along any row; Friedberg Theorem 4.4, §4.2, p. 215)**.** *Let $A \in M_{n\times n}(\mathbb{F})$ and $1 \leq i \leq n$. Then*

$$\det A = \sum_{j=1}^{n} (-1)^{i+j} A_{ij} \det(\tilde{A}_{ij}).$$

Proof. Write row i as $A_{i1}\boldsymbol{e}_1 + \cdots + A_{in}\boldsymbol{e}_n$. Applying Theorem 3.1 to row i repeatedly gives

$$\det A = \sum_{j=1}^{n} A_{ij} \det(B_j),$$

where B_j is obtained from A by replacing row i with $\boldsymbol{e}_j$. By Lemma 3.2,

$$\det(B_j) = (-1)^{i+j} \det(\tilde{A}_{ij}),$$

and the claim follows. □

Remark 3.2 (Extending to columns)**.** Theorem 1 of Lay (§3.1, p. 166) states the stronger fact that expansion may follow any row or any column. A later lecture introduces the identity $\det(A^\top) = \det A$: applying it to $A^\top$, expansion along row j of $A^\top$ is expansion along column j of A, and both give the same number, so the column half holds as well. The theorem of this lecture covers the rows; this remark supplies the column version.

Proposition 3.4 (Determinants of triangular matrices)**.** *Let L be a lower triangular $n \times n$ matrix and U an upper triangular one. Then*

$$\det L = L_{11}L_{22}\cdots L_{nn}, \qquad \det U = U_{11}U_{22}\cdots U_{nn}.$$

In particular the determinant of a diagonal matrix is the product of its diagonal entries, and $\det I_n = 1$.

Proof. Induction on n; for $n = 1$ both sides are that single entry. Let $n \geq 2$. The first row of a lower triangular matrix has only L_{11} nonzero, so by definition $\det L = L_{11} \det(\tilde{L}_{11})$, and $\tilde{L}_{11}$ is again lower triangular of order $n-1$, so the induction hypothesis gives $\det(\tilde{L}_{11}) = L_{22}\cdots L_{nn}$. For an upper triangular matrix, use Theorem 3.3 along the last row: only U_{nn} is nonzero there and its sign is $(-1)^{n+n} = 1$, hence $\det U = U_{nn} \det(\tilde{U}_{nn})$, and $\tilde{U}_{nn}$ is again upper triangular; induction finishes. □

Lemma 3.5 (Two equal or proportional rows)**.** *Let A be an $n \times n$ matrix. If two rows of A are equal, or one is a multiple of another, then* $\det A = 0$.

Proof. Suppose first that two rows are equal. For $n = 2$ those two rows are linearly dependent, so Proposition 2.2 gives $\det A = 0$. For $n \geq 3$ choose a row i that is neither of the two equal rows and expand along it (Theorem 3.3); every cofactor matrix again has two equal rows, so by the induction hypothesis its determinant is 0, hence $\det A = 0$.

Now suppose row s is k times row r. Taking $\boldsymbol{u} = \boldsymbol{0}$ in Theorem 3.1 gives $\det A = k \det A'$, where A' is obtained by replacing row s with row r; the matrix A' has two equal rows, so $\det A' = 0$ and hence $\det A = 0$. □

Theorem 3.6 (How elementary row operations affect the determinant)**.** *Let A be an $n \times n$ matrix. Then*

(a) multiplying a row by k multiplies the determinant by k;
(b) interchanging two rows changes the sign of the determinant;
(c) adding k times one row to another row leaves the determinant unchanged.

Proof. (a) If row r is $\boldsymbol{v}$ and becomes $k\boldsymbol{v}$, take $\boldsymbol{u} = \boldsymbol{0}$ in Theorem 3.1: $\det(\ldots, k\boldsymbol{v}, \ldots) = \det(\ldots, \boldsymbol{0}, \ldots) + k \det(\ldots, \boldsymbol{v}, \ldots)$. A matrix with a zero row has determinant 0, because expanding along that row leaves a factor 0 in every term.

(b) Let B be obtained from A by interchanging rows r and s, and let D be obtained by replacing both rows r and s by $\boldsymbol{a}_r + \boldsymbol{a}_s$. Then D has two equal rows, so $\det D = 0$ by Lemma 3.5. Apply linearity once in row r of D and then once in row s of each resulting matrix:

$$\begin{aligned} 0 = \det D &= \det(\dots, \boldsymbol{a}_r, \dots, \boldsymbol{a}_r + \boldsymbol{a}_s, \dots) + \det(\dots, \boldsymbol{a}_s, \dots, \boldsymbol{a}_r + \boldsymbol{a}_s, \dots) \\ &= \det(\dots, \boldsymbol{a}_r, \dots, \boldsymbol{a}_r, \dots) + \det(\dots, \boldsymbol{a}_r, \dots, \boldsymbol{a}_s, \dots) \\ &\quad + \det(\dots, \boldsymbol{a}_s, \dots, \boldsymbol{a}_r, \dots) + \det(\dots, \boldsymbol{a}_s, \dots, \boldsymbol{a}_s, \dots) = 0 + \det A + \det B + 0, \end{aligned}$$

where the two terms with equal rows vanish; hence $\det B = -\det A$.

(c) Let B be obtained by adding k times row r to row s. Linearity in row s gives $\det B = \det A + k \det C$, where C is A with row s replaced by $\boldsymbol{a}_r$; that matrix has two equal rows, so $\det C = 0$ by Lemma 3.5 and $\det B = \det A$. □

Proposition 3.7 (A matrix of rank less than n has zero determinant)**.** *Let A be an $n \times n$ matrix. If $\operatorname{rank}(A) < n$, then $\det A = 0$.*

Proof. $\operatorname{rank}(A) < n$ means the rows of A are linearly dependent (Lecture 5 notes, Proposition 3.6 and the properties of rank), so some row is a linear combination of the others; say row s is $\sum_{i \neq s} c_i \boldsymbol{a}_i$. Applying Theorem 3.1 repeatedly to row s,

$$\det A = \sum_{i \neq s} c_i \det A_i,$$

where A_i is A with row s replaced by $\boldsymbol{a}_i$. Each A_i has two equal rows (rows i and s), so $\det A_i = 0$ by Lemma 3.5, and hence $\det A = 0$. □

Theorem 3.8 (Multiplicativity of the determinant; Friedberg Theorem 4.7, §4.3, p. 224)**.** *For any $n \times n$ matrices A and B, $\det(AB) = \det A \cdot \det B$.*

Proof. Two cases.

If $\operatorname{rank}(A) < n$, the multiplicative part of Lecture 5 notes, Theorem 3.4 gives $\operatorname{rank}(AB) \leq \operatorname{rank}(A) < n$, so $\det A = \det(AB) = 0$ and both sides agree.

If $\operatorname{rank}(A) = n$ then A is invertible, hence a product of elementary matrices $A = E_m E_{m-1} \cdots E_1$ (Lecture 5 notes, Proposition 3.8). For a single elementary matrix, the effect of left multiplication described in Theorem 3.6 is "multiply by k", "change the sign" or "no change", while $\det E$ is k, -1 or 1 respectively, so $\det(EB) = \det E \cdot \det B$. Applying this successively to $E_m, \dots, E_1$,

$$\det(AB) = \det(E_m \cdots E_1 B) = \det E_m \cdots \det E_1 \cdot \det B = \det A \cdot \det B.$$ □

4 A Worked Example

Example 4.1 (Factor and solve in one pass; written for these notes; method from Lay §2.5, p. 123–127)**.** Let

$$A = \begin{bmatrix} 1 & 2 & 0 & 1 \\ 2 & 5 & 3 & 1 \\ -1 & -1 & 5 & -1 \\ 3 & 4 & -6 & 8 \end{bmatrix}, \qquad \boldsymbol{b} = (0, 4, 9, -5)^{\top}.$$

(1) Using row replacements only, carry A to an echelon form U, and write down L with $A = LU$; (2) solve $A\boldsymbol{x} = \boldsymbol{b}$ with that L and U.

Solution. (1) Subtract twice row 1 from row 2, add row 1 to row 3, subtract three times row 1 from row 4; then subtract row 2 from row 3 and add twice row 2 to row 4:

$$A \sim \begin{bmatrix} 1 & 2 & 0 & 1 \\ 0 & 1 & 3 & -1 \\ 0 & 1 & 5 & 0 \\ 0 & -2 & -6 & 5 \end{bmatrix} \sim U = \begin{bmatrix} 1 & 2 & 0 & 1 \\ 0 & 1 & 3 & -1 \\ 0 & 0 & 2 & 1 \\ 0 & 0 & 0 & 3 \end{bmatrix}.$$

The multipliers are $2, -1, 3$ (first column), $1, -2$ (second column) and 0 (third column); with the sign changed they fill L:

$$L = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 2 & 1 & 0 & 0 \\ -1 & 1 & 1 & 0 \\ 3 & -2 & 0 & 1 \end{bmatrix}, \qquad LU = A.$$

(2) Forward substitution $L\boldsymbol{y} = \boldsymbol{b}$ gives $\boldsymbol{y} = (0, 4, 5, 3)^\top$; back substitution $U\boldsymbol{x} = \boldsymbol{y}$ then gives $x_4 = 1$ from $3x_4 = 3$, $x_3 = 2$ from $2x_3 + x_4 = 5$, $x_2 = -1$ from $x_2 + 3x_3 - x_4 = 4$ and $x_1 = 1$ from $x_1 + 2x_2 + x_4 = 0$. Hence $\boldsymbol{x} = (1, -1, 2, 1)^\top$, and indeed $A\boldsymbol{x} = (0, 4, 9, -5)^\top$. □

Approach. The two questions share one elimination: record the multipliers and change their signs to build L; then solve by one forward and one back substitution, without touching A again.

Example 4.2 (Expanding along the row with the most zeros; written for these notes; method from Friedberg §4.2, p. 215)**.** Find the determinant of

$$A = \begin{bmatrix} 1 & 2 & 0 & 3 \\ 0 & 0 & 5 & 0 \\ 2 & 1 & 1 & 4 \\ 3 & 0 & 0 & 2 \end{bmatrix}.$$

Solution. Row 2 has a single nonzero entry, so expanding along it is cheapest. With $A_{23} = 5$,

$$\det A = (-1)^{2+3} \cdot 5 \cdot \det(\tilde{A}_{23}) = -5 \det \begin{bmatrix} 1 & 2 & 3 \\ 2 & 1 & 4 \\ 3 & 0 & 2 \end{bmatrix}.$$

The remaining determinant of order 3 is again expanded along its row with the most zeros, row 3:

$$\det \begin{bmatrix} 1 & 2 & 3 \\ 2 & 1 & 4 \\ 3 & 0 & 2 \end{bmatrix} = 3 \cdot (-1)^{3+1} \det \begin{bmatrix} 2 & 3 \\ 1 & 4 \end{bmatrix} + 2 \cdot (-1)^{3+3} \det \begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix} = 3 \cdot 5 + 2 \cdot (-3) = 9.$$

Hence $\det A = -5 \cdot 9 = -45$.

Check: expansion along the first row of the same matrix gives the minors

$$\det \begin{bmatrix} 0 & 5 & 0 \\ 1 & 1 & 4 \\ 0 & 0 & 2 \end{bmatrix} = -10, \qquad \det \begin{bmatrix} 0 & 5 & 0 \\ 2 & 1 & 4 \\ 3 & 0 & 2 \end{bmatrix} = 40, \qquad \det \begin{bmatrix} 0 & 0 & 5 \\ 2 & 1 & 1 \\ 3 & 0 & 0 \end{bmatrix} = -15,$$

so $1 \cdot (-10) - 2 \cdot 40 + (-1) \cdot 3 \cdot (-15) = -10 - 80 + 45 = -45$, in agreement with the value above. □

Approach. Count the nonzero entries in each row and expand along the sparsest one; one step takes the 4×4 down to order 3, and the same rule is applied again. The sign depends on the position (i, j): here $(-1)^{2+3} = -1$, $(-1)^{3+1} = +1$ and $(-1)^{3+3} = +1$, and each must be read off separately rather than carried over from the previous row.

Example 4.3 (A determinant with a parameter; written for these notes)**.** Let

$$p(x) = \det \begin{bmatrix} x & 1 & 1 \\ 1 & x & 1 \\ 1 & 1 & x \end{bmatrix}.$$

Find $p(x)$ and solve $p(x) = 0$.

Solution. Expanding along the first row,

$$p(x) = x \det \begin{bmatrix} x & 1 \\ 1 & x \end{bmatrix} - \det \begin{bmatrix} 1 & 1 \\ 1 & x \end{bmatrix} + \det \begin{bmatrix} 1 & x \\ 1 & 1 \end{bmatrix} = x(x^2 - 1) - (x - 1) + (1 - x).$$

Collecting terms,

$$p(x) = x^3 - x - x + 1 + 1 - x = x^3 - 3x + 2 = (x - 1)^2(x + 2),$$

so $p(x) = 0$ at $x = 1$ (double) and $x = -2$. □

Approach. At $x = 1$ the three rows coincide, so the determinant must vanish and $x - 1$ is a factor; dividing it out leaves a quadratic, which is cheaper than solving a cubic from scratch.

5 Checklist and Exercises

By the end of this lecture you should be able to:

1. state the definition of an LU factorization, explain why U need only be an echelon form, and explain why the factorization is unique when no rows have to be interchanged;
2. split $A\boldsymbol{x} = \boldsymbol{b}$ into $L\boldsymbol{y} = \boldsymbol{b}$ and $U\boldsymbol{x} = \boldsymbol{y}$ from $A = LU$, and solve each by the triangular routine;
3. write down the L that belongs to a chain of row replacements, using the multipliers recorded during the elimination, and verify $LU = A$ with one matrix multiplication;
4. state the definition of a permutation matrix and $P^{-1} = P^{\top}$, state $PA = LU$, and explain why the multipliers get reordered by the interchanges that come after them;
5. obtain any column of A^{-1} from one LU factorization, and say which step this saves compared with computing A^{-1} first;
6. state the definition of a determinant of order 2 and explain why it is linear in each row but not linear on matrices as a whole;
7. define the minor and the cofactor, and write the recursive definition of the determinant of order n;
8. explain why expansion along any row gives the same number as expansion along the first row, naming the theorems used;
9. decide the sign of each term in an expansion, and explain why the checkerboard $(-1)^{i+j}$ does not depend on the sign of the entry;
10. pick the sparsest row of a 4×4 matrix and expand along it to reduce the problem to order 3;
11. explain why the determinant of a triangular matrix is the product of its diagonal entries, and say which row or column is used in each of the two cases;
12. state the three effects of elementary row operations on a determinant, and use "record the interchange, record the scaling, and a row replacement costs nothing" to compute a determinant by reducing to an echelon form.

Exercises (printed pages). Lay: §2.5 p. 129 (Exercises 1–6: solving with LU; 7–16: finding an LU factorization), §3.1 p. 167 (Exercises 1–8: expansion along the first row; 9–14: expansion along any row or column), §3.2 p. 176 (Exercises 1–10: row operations and triangular matrices); Friedberg: §4.1 p. 207 (Exercises 1–9: determinants of order 2 and their linearity), §4.2 p. 220

(Exercises 1–14: cofactor expansion and determinants of order n), §4.3 p. 228 (Exercises 1–14: multiplicativity, transpose and invertibility). Each section is preceded by Practice Problems; do those first.

How to use the two texts together:

- **Computation and proof divide cleanly.** How the multipliers are recorded, how a 3×3 is expanded, which row to pick, how the signs are decided: Lay (§2.5, §3.1). Why $PA = LU$ holds, why the determinant can be defined recursively, and why $\det(AB) = \det A \det B$: the theorems of this lecture and Friedberg (§4.2, §4.3).
- The spine of the lecture: *recording elimination as $A = LU$ (or $PA = LU$ when rows must be exchanged) → forward and back substitution → the formula for order 2 → linearity in each row → minors and cofactors → the recursive definition along the first row → expansion along any row → triangular matrices and row operations.*
- The one result worth keeping is Theorem 3.3: it widens "expand along the first row", which is all the definition offers, into "expand along any row", and that is what makes a 4×4 or larger tractable. Its proof uses only linearity in each row (Theorem 3.1) and Lemma 3.2.